🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.AT.7

Simplify Algebraic Expressions

Lesson 6-15 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes (Level 2 Standard)

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can simplify algebraic expressions by combining like terms.

Language Objective: I can explain my steps using the words like terms, combine, simplify, and coefficient.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Like Terms | Combine | Simplify | Coefficient | Term | Distributive property

👆 *Tap any word to see its meaning and a picture.*

1. Terms with the same variable part, like 3y and 8y, are ____________.

2. To add or subtract like terms into a single term is to ____________ them.

3. Rewriting an expression in a shorter form that is still equal to it is to ____________ it.

4. The number multiplied by a variable is the ____________.

5. A number or variable separated by + or – signs is a ____________.

6. Multiplying each part of a sum by a number uses the ______ property.

Write About the Math

Simplify the total order expression. What does each part of the simplified expression tell you?

Simplifica la expresión del pedido total. ¿Qué te dice cada parte de la expresión simplificada?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Like Terms** — *Términos semejantes*
- ☐ **Combine** — *Combinar*
- ☐ **Simplify** — *Simplificar*
- ☐ **Coefficient** — *Coeficiente*
- ☐ **Term** — *Término*

Model: *You can only combine items of the same type, just like like terms.* — Students explain microphones combine with microphones (4 + 2 + 6 = 12) but not with stands, connecting this to combining like terms in algebra.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The total order simplifies to ____m + ____. The ____ tells us ____ and the ____ tells us ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Like Terms
2. **Notes 2:** Combine
3. **Notes 3:** Simplify
4. **Notes 4:** Coefficient
5. **Notes 5:** Term
6. **Notes 6:** Distributive property
7. **Watch for this mistake:** *A common mistake in Simplify Algebraic Expressions is sweeping a constant into the variable terms — for example, simplifying 6x + 4 + 2x as 12x by adding all three numbers (6 + 4 + 2 = 12) instead of recognizing that 4 has no variable and cannot combine with 6x and 2x. The correct simplified expression is 8x + 4 (combine 6x + 2x = 8x, and let the 4 stay on its own). Before you submit, check that every term you combined has the exact same variable part.*
8. **Try It 1:** A. 4y — *Subtract the coefficients of the like terms: 7 − 3 = 4. Keep the variable y, giving 4y.*
9. **Try It 2:** A. 9m + 4 — *6m + 3m = 9m. The constant 4 has no like term, so it stays. Result: 9m + 4.*